

Executive Summary

This material covers the fundamental principles of cellular biology, including the structure and function of cells, photosynthesis, cellular respiration, cell transport, DNA and protein synthesis, cell division, genetics, enzymes, homeostasis, and human body systems. It provides a comprehensive overview of key biological processes and concepts essential for understanding living organisms.

KEY TAKEAWAYS

The cell is the basic structural and functional unit of all living organisms, and every organism is composed of one or more cells.

Photosynthesis transforms light energy into chemical energy in green plants, with the summary equation $6\text{CO}_2 + 6\text{H}_2\text{O} = \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$.

Cellular respiration consists of glycolysis, the Krebs cycle, and the electron transport chain, with the summary equation $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 = 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{ATP}$.

DNA carries genetic instructions, and its replication, transcription, and translation are essential for protein synthesis.

Mitosis produces two genetically identical daughter cells, while meiosis produces gametes with genetic variation.

Genes and alleles determine traits, and Punnett squares predict inheritance patterns.

Enzymes act as biological catalysts, with active sites that bind substrates and optimal temperatures for function.

Homeostasis maintains a stable internal environment through mechanisms like negative and positive feedback.

The human circulatory, respiratory, nervous, and endocrine systems work together to maintain bodily functions.

Topics

TOPIC 1

Cellular Biology

The cell is the basic structural and functional unit of all living organisms. Every organism is composed of one or more cells, and the cell theory states that all living things are composed of cells and new cells arise from existing cells.

Cell Theory

All living things are composed of cells.

New cells are produced from existing cells.

Cell Structure

The nucleus is a membrane-bound organelle that contains genetic material.

Mitochondria are organelles responsible for producing ATP through cellular respiration.

TOPIC 2

Photosynthesis

Photosynthesis is the process by which green plants transform light energy into chemical energy. It occurs in the chloroplast, which contains the green pigment chlorophyll.

Process Overview

Photosynthesis transforms light energy into chemical energy.

Occurs in the chloroplast, which contains chlorophyll.

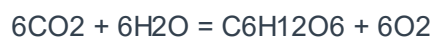
Key Components

Chlorophyll absorbs light energy.

Light-dependent reactions occur in the thylakoid membranes and produce ATP and NADPH.

The Calvin cycle fixes carbon dioxide into glucose.

Summary Equation



TOPIC 3

Cellular Respiration

Cellular respiration consists of three main stages: glycolysis, the Krebs cycle, and the electron transport chain. It generates ATP, the energy currency of the cell.

Stages of Respiration

Glycolysis occurs in the cytoplasm and breaks down glucose into two molecules of pyruvate.

The Krebs cycle occurs in the mitochondrial matrix.

The electron transport chain occurs on the inner mitochondrial membrane.

Anaerobic Process

Fermentation regenerates NAD⁺ when oxygen is unavailable.

Summary Equation



TOPIC 4

Cell Transport

The cell membrane is a selectively permeable barrier that controls what enters and leaves the cell.

Transport mechanisms include diffusion, osmosis, active transport, and endocytosis.

Passive Transport

Diffusion is the passive movement of molecules from an area of high concentration to low concentration.

Osmosis is the diffusion of water across a selectively permeable membrane.

Active Transport

Active transport moves substances against their concentration gradient, requiring energy in the form of ATP.

Endocytosis

Endocytosis is the process by which the cell engulfs large particles by wrapping the membrane around them.

TOPIC 5

DNA and Protein Synthesis

DNA carries genetic instructions for development and functioning. Protein synthesis involves DNA replication, transcription, and translation.

DNA

DNA is the molecule that carries genetic instructions for living organisms.

DNA Replication

DNA replication produces two identical DNA molecules from one double-stranded molecule.

Transcription

Transcription copies the DNA sequence of a gene into messenger RNA.

Translation

Translation is the process by which the ribosome builds a protein from the mRNA sequence.

Each three-nucleotide codon on mRNA specifies one amino acid.

Genetic Code

The genetic code is the set of rules by which information encoded in genetic material is translated into proteins.

TOPIC 6

Cell Division

Cell division involves mitosis and meiosis. Mitosis produces two genetically identical daughter cells, while meiosis produces gametes with genetic variation.

Mitosis

Mitosis is the process of nuclear division that produces two genetically identical daughter cells.

Cell Cycle

The cell cycle consists of interphase, mitosis, and cytokinesis.

Stages of Mitosis

Interphase: Cell grows, copies DNA, and prepares for division.

Prophase: Chromosomes condense.

Anaphase: Sister chromatids are pulled apart to opposite poles of the cell.

TOPIC 7

Genetics

Genetics is the study of heredity, including genes, alleles, genotypes, and phenotypes. Punnett squares predict inheritance patterns.

Genes and Alleles

A gene is the basic unit of heredity, a segment of DNA that codes for a specific trait.

An allele is an alternative form of a gene at the same locus.

Genotype and Phenotype

The genotype is the genetic makeup of an organism.

The phenotype is the observable characteristics of an organism.

Dominant and Recessive Alleles

A dominant allele is expressed when only one copy is present.

A recessive allele is only expressed when two copies are present.

Punnett Squares

Punnett squares are used to predict the probability of offspring inheriting particular traits.

TOPIC 8

Enzymes

Enzymes are proteins that act as biological catalysts, speeding up chemical reactions without being consumed. They have active sites that bind substrates and optimal temperatures for function.

Enzyme Function

An enzyme is a protein that acts as a biological catalyst, speeding up chemical reactions without being consumed.

Active Site

The active site is the region of an enzyme where the substrate binds.

Induced Fit Model

The induced fit model describes how the enzyme changes shape to accommodate the substrate.

Optimal Conditions

Optimal temperature is the temperature at which an enzyme works fastest.

Denaturation

Denaturation is the loss of an enzyme's three-dimensional shape, which destroys its function.

Homeostasis

Homeostasis is the maintenance of a stable internal environment despite external changes. It involves mechanisms like negative and positive feedback to regulate bodily functions.

Mechanisms

Negative feedback reverses a change, bringing the system back to its set point.

Positive feedback amplifies a change and drives the system further from its set point.

Examples

Normal human body temperature is maintained at approximately 37 degrees Celsius.

Blood glucose concentration is regulated by the hormones insulin and glucagon.

Insulin is secreted when blood glucose is high, promoting glucose uptake.

Human Systems

The human body consists of multiple systems, including the circulatory, respiratory, nervous, and endocrine systems, which work together to maintain bodily functions.

Circulatory System

The circulatory system transports oxygen, nutrients, and waste products throughout the body.

The heart is the muscular organ that pumps blood through the circulatory system.

Respiratory System

The respiratory system is responsible for gas exchange, bringing oxygen into the body and removing carbon dioxide.

Nervous System

The nervous system coordinates rapid responses through electrical signals called nerve impulses.

Endocrine System

The endocrine system regulates long-term processes through chemical messengers called hormones.

Terms & Definitions

Term	Definition
Cell	The basic structural and functional unit of all living organisms. Every organism is composed of one or more cells.
Cell theory	A fundamental principle of biology that states that all living things are composed of cells and that new cells are produced from existing cells.
Nucleus	The membrane-bound organelle that contains the genetic material.
Mitochondria	The organelle responsible for producing ATP through cellular respiration.
Photosynthesis	The process by which green plants transform light energy into chemical energy.
Chloroplast	The organelle where photosynthesis occurs and contains the green pigment chlorophyll.
Chlorophyll	The pigment that absorbs light energy during photosynthesis.
Light-dependent reactions	Reactions that occur in the thylakoid membranes and produce ATP and NADPH.
Calvin cycle	The light-independent stage of photosynthesis that fixes carbon dioxide into glucose.
Cellular respiration	A process consisting of three main stages: glycolysis, the Krebs cycle, and the electron transport chain.
Glycolysis	The first stage of cellular respiration that occurs in the cytoplasm and breaks down glucose into two molecules of pyruvate.
Krebs cycle	The stage of cellular respiration that occurs in the mitochondrial matrix.
Electron transport chain	The final stage of cellular respiration that occurs on the inner mitochondrial membrane.
Fermentation	The anaerobic process that regenerates NAD ⁺ when oxygen is unavailable.
Cell membrane	The selectively permeable barrier that controls what enters and leaves the cell.
Diffusion	The passive movement of molecules from an area of high concentration to low concentration.
Osmosis	The diffusion of water across a selectively permeable membrane.
Active transport	The movement of substances against their concentration gradient, requiring energy in the form of ATP.
Endocytosis	The process by which the cell engulfs large particles by wrapping the membrane around them.
DNA	The molecule that carries the genetic instructions for the development and functioning of living organisms.

Term	Definition
DNA replication	The process by which a double-stranded DNA molecule is copied to produce two identical DNA molecules.
Transcription	The process by which the DNA sequence of a gene is copied into messenger RNA.
Translation	The process by which the ribosome builds a protein from the mRNA sequence.
Genetic code	The set of rules by which information encoded in genetic material is translated into proteins.
Mitosis	The process of nuclear division that produces two genetically identical daughter cells.
Cell cycle	Consists of interphase, mitosis, and cytokinesis.
Interphase	The stage during which the cell grows, copies its DNA, and prepares for division.
Prophase	The first stage of mitosis, during which chromosomes condense.
Anaphase	The stage during which sister chromatids are pulled apart to opposite poles of the cell.
Gene	The basic unit of heredity, a segment of DNA that codes for a specific trait.
Allele	An alternative form of a gene at the same locus.
Genotype	The genetic makeup of an organism.
Phenotype	The observable characteristics of an organism.
Dominant allele	An allele that is expressed when only one copy is present.
Recessive allele	An allele that is only expressed when two copies are present.
Punnett square	A tool used to predict the probability of offspring inheriting particular traits.
Enzyme	A protein that acts as a biological catalyst, speeding up chemical reactions without being consumed.
Active site	The region of an enzyme where the substrate binds.
Induced fit model	Describes how the enzyme changes shape to accommodate the substrate.
Optimal temperature	The temperature at which an enzyme works fastest.
Denaturation	The loss of an enzyme's three-dimensional shape, which destroys its function.
Homeostasis	The maintenance of a stable internal environment despite external changes.

Term	Definition
Negative feedback	A mechanism that reverses a change, bringing the system back to its set point.
Positive feedback	A mechanism that amplifies a change and drives the system further from its set point.
Insulin	A hormone secreted when blood glucose is high, promoting glucose uptake.
Glucagon	A hormone that regulates blood glucose concentration.
Circulatory system	The system that transports oxygen, nutrients, and waste products throughout the body.
Heart	The muscular organ that pumps blood through the circulatory system.
Respiratory system	The system responsible for gas exchange, bringing oxygen into the body and removing carbon dioxide.
Nervous system	The system that coordinates rapid responses through electrical signals called nerve impulses.
Endocrine system	The system that regulates long-term processes through chemical messengers called hormones.
C₆H₁₂O₆	The summary equation of photosynthesis is $6\text{CO}_2 + 6\text{H}_2\text{O} = \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$.
$6\text{CO}_2 + 6\text{H}_2\text{O} = \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$	The summary equation of photosynthesis.
$\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 = 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{ATP}$	The summary equation of cellular respiration.
NADPH	A product of the light-dependent reactions in photosynthesis.
NAD⁺	A molecule regenerated during fermentation when oxygen is unavailable.
mRNA	Messenger RNA, which carries the genetic information from DNA to the ribosome for protein synthesis.
Codon	A three-nucleotide sequence on mRNA that specifies one amino acid.
Chromosomes	Condensed structures of genetic material visible during cell division.
Sister chromatids	Identical copies of a chromosome formed during DNA replication.
Cytokinesis	The process by which the cytoplasm divides, resulting in two daughter cells.

Key Facts & Formulas

Formula / Fact	Context
respiration is $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 = 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{ATP}$. $\text{C}_6\text{H}_{12}\text{O}_6$	respiration is $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 = 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{ATP}$. oduce glucose and release oxygen. The summary equation of photosynthesis is $6\text{CO}_2 + 6\text{H}_2\text{O} = \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$. The light-dependent reactions occur in the thylakoid membranes and produce ATP and